

DSC EnvisaLink Ethernet Driver

Overview

This document describes the setup of, and the communication protocol implemented in the Adroit protocol driver, with the following logic unit: DSC EnvisaLink.

Device Configuration

DSC Envisalink Driver : DSC1

Configuration selection

☐ Enable Secondary

☒ Primary

☐ Secondary

IP Settings

IP Address: 10 . 10 . 10 . 121

Passwords

TPI Password: xxxx

DSC Master Code: xxxx

DSC Installer Code: xxxx

Internal Settings

Keep alive poll frequency (ms): 4000

Status poll frequency (ms): 60000

☒ Set alarm system time from SCADA

Time update frequency (m): 10

System timeout (ms): 5000

OK Cancel

Configuration Selection

Enable Secondary – Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

This setting is disabled for this driver.

IP Settings

IP Address – The IP Address that will be used for communicating with the controller.

Passwords

Please note. Passwords are encrypted using 16 bit key encryption before being stored in the registry.

TPI Password – Enter your envisalink webpage password here

DSC Master Code – Enter the DSC alarm panel's master code. This is used to arm and disarm the alarm panel.

DSC Installer Code – Enter the DSC alarm panel's installer code. This is used when configuring the alarm panel.

Internal Settings

Note. This driver has an internal thread which handles the scanning. Changing these settings, change the frequency in which the scanning thread adds the following to the read queue.

Keep alive poll frequency (ms) – This is the time delay between keep alive polls. This value is measured in milliseconds and must be between 1000 and 10000.

Status poll frequency (ms) – This is the time delay between status polls. This value is measured in milliseconds and has a minimum of 10000ms.

Set alarm system time from SCADA – If this checkbox is checked, the alarms panel's system time will be set with the time from the Adroit agent server.

Time update frequency (m) – The time on the alarm panel may drift, if "Set alarm system time from SCADA" is check, the alarm panel's system time will be updated with this frequency.

Timeout (ms) – If the driver does not receive a valid response from the device within this period (measured in milliseconds), the port will be closed. Note. This value must be greater than the "keep alive poll period" in the "Internal Settings" section.

Wiring

Standard Ethernet connection

Supported addresses

The DSC EnvisaLink Adroit Ethernet driver supports the following data address types:

Address	Description	Min Addr	Max Addr	Addressing example	Supported Adroit Data types	Read/Write
PAR	Partition status request	1	8	PAR 1 PAR 8	Integer	Read
ZON	Zone status request	1	64	ZON 1 ZON 64	Integer	Read
ALM	Alarm status request	1	5	ALM 1 ALM 5	Digital Integer	Read
GEN	General status request	1	4	GEN 1 GEN 4	Digital Integer	Read
TBL	Trouble LED request	1	8	TBL 1 TBL 8	Digital Integer	Read
PAD	Partition Arm/Disarm	1	8	PAD 1 PAD 8	Integer	Write
KEY	Keypad Sequence	1	8	KEY 1 KEY 8	String	Write
PAA	Panic Alarm Activate	0	0	PAA	Integer	Write
PGM	Programmable output activate	1	4	PGM1 PGM4	Integer	Write

Note. Please read the Address Descriptions section before using these addresses!

Address Descriptions

The addresses are either read only or write only. Read only addresses are sent from the driver to adroit in an unsolicited manner. Write only addresses will be cleared (to 0 if digital or integer and to "" if string) once the write is complete.

Partition status request (PAR)

This is a read only address. The value sent to adroit when scanning this address, represents the state of the partition being scanned.

Value	State
1	Partition ready
2	Partition not ready
4	Partition armed in away mode
8	Partition armed in stay mode
16	Partition armed in zero entry away mode
32	Partition armed in zero entry stay mode
64	Partition in alarm
128	Partition in exit delay
256	Partition in entry delay

Zone status request (ZON)

This is a read only address. The value sent to adroit when scanning this address, represents the state of the zone being scanned.

Value	State
1	Zone restored
2	Zone Open
4	Zone Fault
8	Zone Tamper
16	Zone Alarm

Alarm status request (ALM)

Each address references a different type of alarm. The following table indicates what type of alarm each addresses refers to.

Address	Alarm Type
ALM 1	Duress alarm state
ALM 2	Fire alarm state
ALM 3	Auxiliary key alarm state
ALM 4	Panic alarm state
ALM 5	Smoke alarm state

This is a read only address. The value sent to adroit when scanning this address, represents the state of the alarm being scanned.

Value	State
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0	Not alarmed
1	In Alarm

General status request (GEN)

This is a read only address. Each address references a unique general status. The table below indicates what each address references to.

Address	Status Type
GEN 1	Panel battery trouble, battery low or battery has been removed.
GEN 2	Panel AC trouble, AC power to the panel has been removed.
GEN 3	System bell trouble, an open circuit has been detected across the bell terminals.
GEN 4	General system tamper.

The value sent to adroit when scanning this addresses, represents the general state of the status type being scanned.

Value	State
0	Good status
1	Bad status

Trouble LED request (TBL)

This is a read only address. The table below indicates the state of the trouble LED on the keypad of the partition being scanned (1-8).

Value	State
0	LED off
1	LED on

Partition Arm/Disarm (PAD)

This is a write only address. The table below describes the resultant action on the partition scanned (1-8), when writing the following values from adroit.

Write Value	Result of write
1	Arm partition in away mode
2	Arm partition in stay mode
3	Arm partition with zero entry delay
4	Disarm partition with user code entered in the dialog

Keypad Sequence (KEY)

This is a write only address. This address allows the driver to simulate a keystroke sequence as if it were being pressed on a specific partition's (1-8) keypad. Only the following characters can be sent '1 2 3 4 5 6 7 8 9 * #'. If any other chars are found in the character sequence, the write will fail.

Panic Alarm Activate (PAA)

This is a write only address. The following table indicates which panic alarms will be triggered when certain values are written to this address.

Write Value	Result of write
1	Trigger fire alarm
2	Trigger ambulance alarm
3	Trigger police alarm

Programmable output activate (PGM)

This is a write only address. The address index (1-4) represents the PGM that will be outputted when written. Valid write values are 1-8. The write value is partitions which activates the programmable output.

Message Protocol Description

DSC EnvisaLink message structure

Command Code	Data	Checksum	Carriage-return linefeed
CCC	DDD...DDD	KK	\r\n

Command Code:

This three digit code when sent to the device issues a command, i.e. tells the device what to do. However when receiving a message from the device, this code tells the driver what type of information is being received.

Data:

This part of the message is variable length depending on the command code sent, and contains the data required for the command code.

Checksum:

The checksum is two characters wide. The checksum is calculated by adding the hex value of all command and data digits, and truncating the result to 8 bits.

Carriage-return-linefeed:

The carriage-return and linefeed characters are appended to the end of each message.

Revision information

Date:	Changes:
10/02/2012	First Release notes!

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